Capstone Group 19

Progress Report 2: Design Inputs

An Adaptive Wrist-Worn Device That Learns Each Patient's Tremor Pattern to Suppress and Track Parkinson's Disease Rest Tremor in Daily Life

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1 Updated User and Problem Framing

1.1 Changes Since Report 1

Adaptive tremor suppression remains an active longer-term objective. Following PR1 feedback, immediate work prioritizes monitoring Parkinson's rest tremor, with posture and activity analysed separately. Parkinson Canada remains the stakeholder. FDA evidence informs measurement and coverage reporting but does not establish suppression feasibility.

1.2 User Priorities and Intended Benefit

The proposed daily timeline will summarize tremor fraction, detected frequency, recording coverage and user-entered context, supported by guided rest/posture measurements. Analysis will use non-overlapping three-second windows. Valid windows require complete, correctly timed, unclipped on-wrist data; these quality checks remain unverified. Detected tremor means a valid window passing the provisional signal gate, not a clinical diagnosis. Within each labelled context, tremor fraction is detected-window duration divided by valid-window duration. Valid gate-negative windows count as no detected tremor; invalid windows are excluded and flagged. Coverage is valid time divided by scheduled recording time. Frequency is displayed only for detected windows. Patients and clinicians still need to confirm useful summaries and recording burden.

Need statement: Patients and clinicians need an understandable record of tremor patterns over time, supported by comparable guided measurements, to discuss symptom changes.

2 Design Inputs

2.1 Requirements and Traceability

Table 1 links needs to functional, performance, interface and physical inputs, metrics and verification. Binary criteria are proposed; unsupported values remain TBD. Accuracy and coverage take priority. Need IDs: N4 safe use; N5 comfort; N6 useful trends; N7 affordability; N8 control of personal data. Acceptance limits will be justified from the intended interpretation, reference-measurement uncertainty and user requirements before formal verification.

Table 1. Design inputs and their evidence

ID	Input and measurable criterion	Need or source	Verification
DI-01 Functional	Each summary includes time, estimated tremor fraction, frequency, context and coverage. Unavailable values are labelled; frequency is shown only when validity and detection checks pass.	N6; R-01; team proposal	Replay labelled intervals and check displayed values, missing-data states and detection status.
DI-02 Performance	Frequency error (Hz), detection sensitivity and false-positive rate, evaluated by task; test conditions and acceptance limits TBD.	N6; team proposal, informed by [1]	Use known-frequency signals and independently labelled recordings, including ordinary movement and no-tremor cases.
DI-03 Performance	100% injected gaps flagged; valid-time denominator excludes every gap.	N6; R-01; informed by [2]	Known-gap replay; compare computed coverage.

ID	Input and measurable criterion	Need or source	Verification
DI-04 Interface	100% guided comparisons retain task/wrist labels; no silent pooling.	N6; R-01; team proposal	Audit labelled rest/posture comparisons.
DI-05 Physical	Mass (g), wear duration (h) and discomfort score: scale/limits TBD.	N4/N5; R-02; user review pending	Measure mass; define the discomfort scale and wear protocol with intended users before evaluation.

2.2 Constraints and Trade-Offs

Budget, hardware and wear-time limits remain unagreed. Compare concepts using estimated parts cost (CAD), recording time between charges (h), setup time (min), daily user actions and replaceability of the strap and battery. Evaluate sampling versus battery use and context detail versus interaction burden. Assess one-handed operation, caregiver assistance and electronic waste during concept comparison. Privacy input (N8): sharing must require an explicit user action; verify with a no-consent export test.

3 Concept Development Process and Preliminary Analysis

3.1 Process and Supervisor Consultation

Generate alternatives by combining existing research loggers or custom sensing with passive timelines or guided-task workflows. Screen each against safety and data-integrity inputs; compare survivors using common data-quality, usability, cost, integration, privacy and repairability criteria. Agree weights with supervisors, record evidence and uncertainty, and check whether plausible weight changes reverse rankings. Final selection will follow concept comparison and supervisor review.

Supervisor agreement on concept generation, screening criteria and the comparison method remains pending.

3.2 Feasibility Evidence and Uncertainty

Cala ONE missed its primary spiral endpoint ($p = .263$), but received De Novo authorization following benefit-risk review [3]. Cala kIQ's PD indication concerns postural/kinetic hand tremor [1]. Felix's ET review emphasized missing-data analysis [2, p. 21]. These findings motivate task-specific evaluation, prespecified outcomes and recording coverage; they do not establish our monitor's performance.

Inherited PADS tables and code are available, but raw-data recovery and reproduction remain pending. Its short clinic tasks [4] can support guided-task evaluation but cannot alone validate daily trends.

An October 23 host test of the unchanged features.c code used 624 samples at 208 Hz per window. Signed, noise-free sinusoids at 4, 5, 6, 4.25 and 5.75 Hz, each at three phases and 20 degrees/s peak amplitude, produced 15 cases. Frequency error was zero for the nine grid-aligned cases and 0.25 Hz for the six between-bin cases; the gate accepted all 15. It rejected zero-input and 1.2 Hz sine controls, although both returned a nominal 3 Hz estimate. These function-level results support detection-gated display; they do not validate sensor preprocessing, ordinary-motion rejection or clinical accuracy. Next, test noisy and mixed-motion inputs and target-hardware timing.

Suppression work will next compare candidate mechanisms using literature and a simplified wrist-response model, examining tremor reduction versus interference with voluntary movement. These suppression comparisons remain planned.

4 Preliminary Risk Analysis

4.1 Hazards and Design Implications

Table 2 links monitoring hazards to proposed, unvalidated controls. For suppression, unintended actuation could disturb voluntary movement or injure the wrist. Compare mechanisms by controllability and safe operating limits; verify a safe stop and mechanical or electrical limits before any wearer testing. Suppression-specific controls remain to be developed. Manufacturing and deployment risks will be refined as concepts become concrete.

Table 2. Preliminary risks and their implications for design

Risk and context	Potential consequence	Control or evidence needed	Effect on concepts
R-01 Misleading timeline	Missing data, task changes or false detections could imply symptom change and cause distress or inappropriate self-management.	Show coverage and context; suppress unsupported frequency values; verify missing-data and no-tremor cases.	Prioritize interpretability; avoid diagnostic claims.
R-02 Wear and power	Strap pressure, skin irritation or battery heating could harm a wearer during prototyping or use.	Inspect power protection and enclosure integrity on the bench; review contact materials and wear limits before participant testing.	Compare fit, contact materials and safe operating limits.

4.2 Standards and Ethical Considerations

ISO 14971:2019 is a proposed framework for hazard identification, controls and residual-risk review [5]; compliance has not been established. Confirm research ethics and consent requirements under TCPS 2 [6]. Assess the applicability of Canada's Medical Devices Regulations and any investigational-testing authorization requirements using the intended use and device classification [7]. U.S. stimulator clearances do not establish Canadian authorization for this monitor. Unresolved requirements may delay participant testing. Restrict access to identifiable data, define retention and deletion, and exclude diagnostic, staging and medication-advice claims.

References

[1] U.S. Food and Drug Administration, ‘Cala kIQ - 510(k) summary’, U.S. Food and Drug Administration, Nov. 2022. [Online]. Available: https://www.accessdata.fda.gov/cdrh_docs/pdf22/K222237.pdf

[2] U.S. Food and Drug Administration, ‘Felix NeuroAI - 510(k) summary’, U.S. Food and Drug Administration, Jul. 2025. [Online]. Available: https://www.accessdata.fda.gov/cdrh_docs/pdf25/K250096.pdf

[3] U.S. Food and Drug Administration, ‘Cala ONE - De Novo classification review’, U.S. Food and Drug Administration, Apr. 2018. [Online]. Available: https://www.accessdata.fda.gov/cdrh_docs/reviews/DEN170028.pdf

[4] J. Varghese, A. Brenner, L. Plagwitz, C. van Alen, M. Fujarski, and T. Warnecke, ‘PADS - Parkinson’s Disease Smartwatch dataset (version 1.0.0)’. PhysioNet, Mar. 25, 2024. doi: 10.13026/m0w9-zx22.

[5] International Organization for Standardization, ‘ISO 14971:2019 - Medical devices: Application of risk management to medical devices’, public standard description, 2019. Accessed: Oct. 23, 2026. [Online]. Available: https://www.iso.org/standard/72704.html

[6] Canadian Institutes of Health Research, Natural Sciences and Engineering Research Council of Canada, and Social Sciences and Humanities Research Council of Canada, ‘Tri-Council Policy Statement: Ethical Conduct for Research Involving Humans - TCPS 2 (2022)’, 2022. [Online]. Available: https://ethics.gc.ca/eng/documents/tcps2-2022-en.pdf

[7] Government of Canada, ‘Medical Devices Regulations’, SOR/98-282, Part 3, s. 80. Accessed: Oct. 23, 2026. [Online]. Available: https://laws-lois.justice.gc.ca/eng/regulations/SOR-98-282/section-80.html